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ENDGAMES

STATISTICAL QUESTION

Intention to treat analysis versus per protocol analysis of trial data

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Researchers evaluated the effectiveness of a self management programme for arthritis on the overall function of patients with osteoarthritis in primary care. A randomised controlled trial study design was used. The intervention was attendance at six sessions of self management of arthritis, plus an education booklet. The control group received the education booklet only. Participants were patients aged 50 years or more who had osteoarthritis of the hips or knees (or both) and pain or disability (or both). In total, 812 patients were recruited and randomised to the intervention (n=406) or control (n=406).[1]

The primary outcome was quality of life, as assessed by the short form health survey (SF-36). Secondary outcomes included physical and psychosocial measures. Outcome measures were recorded by postal questionnaires, collected at baseline and 12 months. Analysis was performed on an intention to treat approach. The researchers reported that the intervention group showed a significant reduction on the anxiety subscore of the hospital anxiety and depression scale at 12 months (mean difference 0.62, 95% confidence interval 0.16 to 1.08). The intervention group also showed a significant improvement on the arthritis self efficacy scale for pain (0.98, 0.07 to 1.89) and self efficacy for other aspects of management (1.58, 0.25 to 2.90). Per protocol analysis produced similar results to the intention to treat analysis with respect to significant findings. It was concluded that the self management of arthritis programme reduced anxiety and improved participants' perceived self efficacy to manage symptoms, although it had no significant effect on pain, physical functioning, or contact with primary care.

Which of the following statements, if any, are true?

a) Trial participants who did not start their allocated treatment were excluded from the intention to treat analysis

b) Intention to treat analysis maintained the original group composition achieved by randomisation

c) Intention to treat analysis provides a pragmatic estimate of the benefit of intervention

d) The per protocol analysis included only those participants who completed the protocol for their allocated treatment.

Answers

Statements *b*, *c*, and *d* are true, whereas *a* is false.

The effectiveness of the self management programme for arthritis on the overall function of patients with osteoarthritis in primary care was investigated. A randomised controlled trial was used. The principle of intention to treat was used to analyse the data. This meant that all participants recruited to the trial were included in the analysis and compared in the outcome measures on the basis of the treatment group to which they were originally randomly allocated. This was regardless of whether they started the treatment allocated (*a* is false), subsequently withdrew from the trial, deviated from the treatment protocol, received a different treatment, or were lost to follow-up—for example, because they died.

Intention to treat analysis maintained the original treatment group composition achieved after the random allocation of the trial participants (*b* is true). Therefore, confounding between the treatment groups was minimised. Confounding factors are those that can influence treatment and outcome measures, such as demographics, prognostic factors, and other characteristics that might influence someone to participate in, or withdraw from, a trial. Therefore, any differences between the treatment groups at the end of the study will have been the result of differences in treatment received and not confounding between treatment groups at baseline. In general, confounding is reduced as the sample size increases.

Intention to treat analysis is a pragmatic approach—it reflects what would happen in clinical practice (*c* is true). Patients may not start, complete, or continue with their prescribed treatment. Sometimes treatments and interventions are not acceptable or well tolerated. Therefore, intention to treat analysis provides an assessment of the benefits of treatment in clinical practice. The above study was designed as a pragmatic primary care based trial.[2] Intention to treat analysis is probably more often used in

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pragmatic trials, which aim to measure the effectiveness of an intervention in routine practice.

In the above trial, the intention to treat analysis depended on all trial participants providing measurements of the outcome variables. However, as is typical of clinical trials it was inevitable that data were missing for some participants. The researchers reported that after recruitment to the trial and randomisation to treatment group, 57 (7.1%) participants withdrew from the trial, 128 (15.8%) did not respond to the invitation to participate, and seven (0.9%) died. Furthermore, the length of trial follow-up was one year. It was therefore inevitable that not all participants in the intervention group were able to attend all of the six sessions of self management. In addition, not all participants would have returned their questionnaires, which provided measurements of the outcome variables. To perform the intention to treat analysis the researchers estimated the missing data using complex methodology. The imputed data predicted the outcomes for those participants with missing data as if they had completed the trial. Estimating missing data in clinical trials has been described in a previous question.[3]

The researchers also performed a per protocol analysis that included only those participants who completed the protocol for the treatment that they were originally allocated to (*d* is true). Patients were considered to have complied with their treatment protocol if they completed and returned all questionnaires, thereby providing a full set of data on the outcome measures, and participants who were allocated to the intervention needed to have attended four or more sessions. Attendance at four sessions was considered clinically to be the minimal "dose" of the intervention. Participants were included in the analysis and the outcome measures compared on the basis of the treatment group to which they were originally allocated. By including only those participants who complied with the trial protocol, the per protocol analysis reflected the effects of the intervention unaffected by, for example, protocol deviations or non-adherence. A per protocol analysis is more often used in explanatory trials that aim to measure the effects of an intervention under ideal or experimental conditions.[2]

A per protocol analysis for a trial may be biased because participants may be excluded after randomisation if they did not complete the protocol of the allocated treatment. This might mean that the original comparability of the treatment groups in baseline characteristics achieved by randomisation had not been maintained. This would result in confounding. Therefore, any differences between treatment groups at the end of the study might not have been due to differences in treatment received but be a result of differences between treatment groups in their baseline characteristics.

The researchers reported that about 30% of the intervention group did not attend any "challenging arthritis" sessions. Nonetheless, the results for the intention to treat and per protocol analyses were similar—that is, there was little difference in the estimated effects of the intervention for all participants recruited to the trial compared with those who attended four or more sessions. Both sets of analyses indicated that the intervention significantly reduced the anxiety subscore of the hospital anxiety and depression scale at 12 months and improved the arthritis self efficacy scales for pain and for other aspects of management. It was suggested that poor attendance or not receiving the suggested clinically significant dose of the intervention had a limited impact on the final results.

The CONSORT (Consolidated Standards of Reporting Trials) statement is a set of guidelines that was established to improve the quality of reporting of clinical trials—in particular, alleviating problems arising from inadequate reporting. The guidelines recommend intention to treat analysis as standard practice. Typically, a per protocol analysis will also be performed alongside an intention to treat approach to enable the influence of any missing data to be investigated.

Competing interests: None declared.

1 Buszewicz M, Rait G, Griffin M, Nazareth I, Patel A, Atkinson A, et al. Self management of arthritis in primary care: randomised controlled trial. *BMJ* 2006;333:879.
2 Sedgwick P. Explanatory trials versus pragmatic trials. *BMJ* 2014;349:g6694.
3 Marston L, Sedgwick P. Randomised controlled trials: missing data. *BMJ* 2014;349:g4656.

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